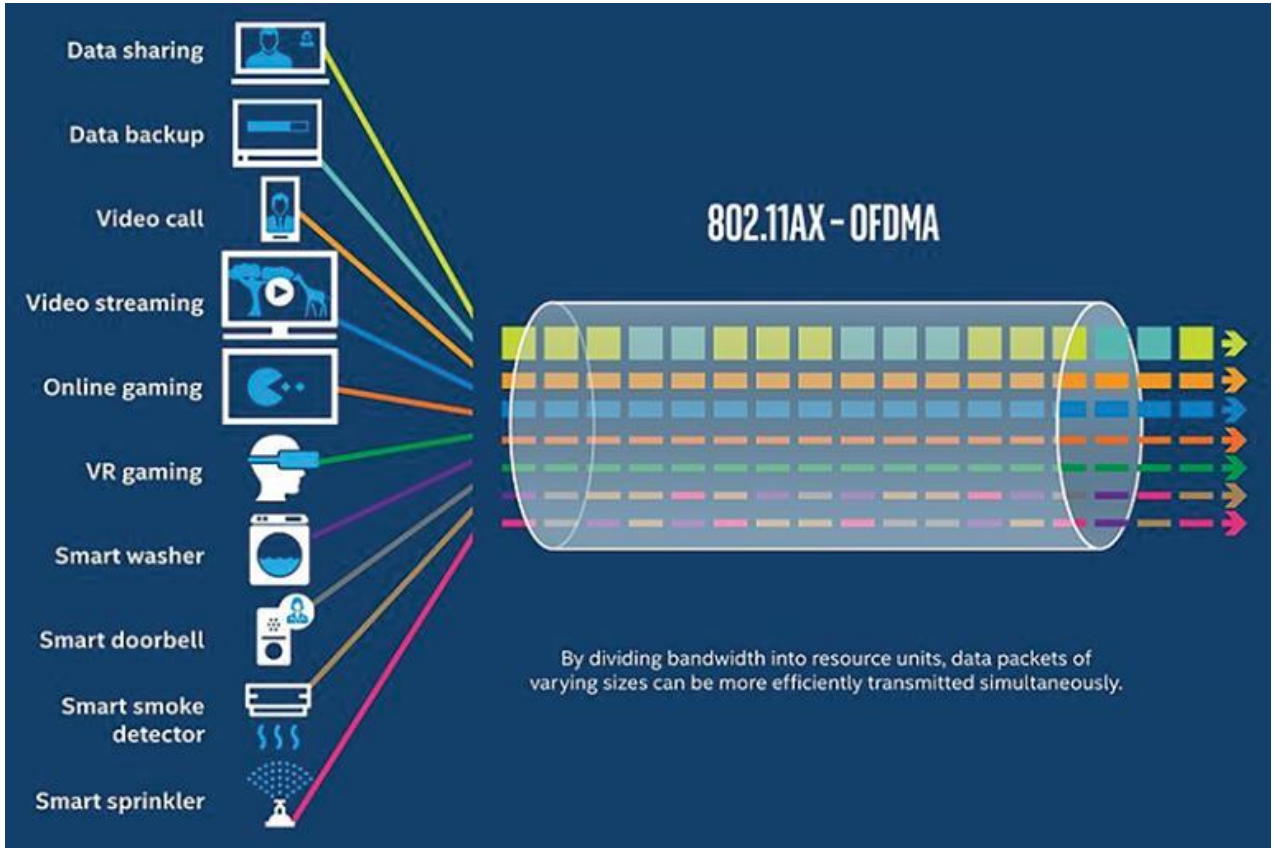
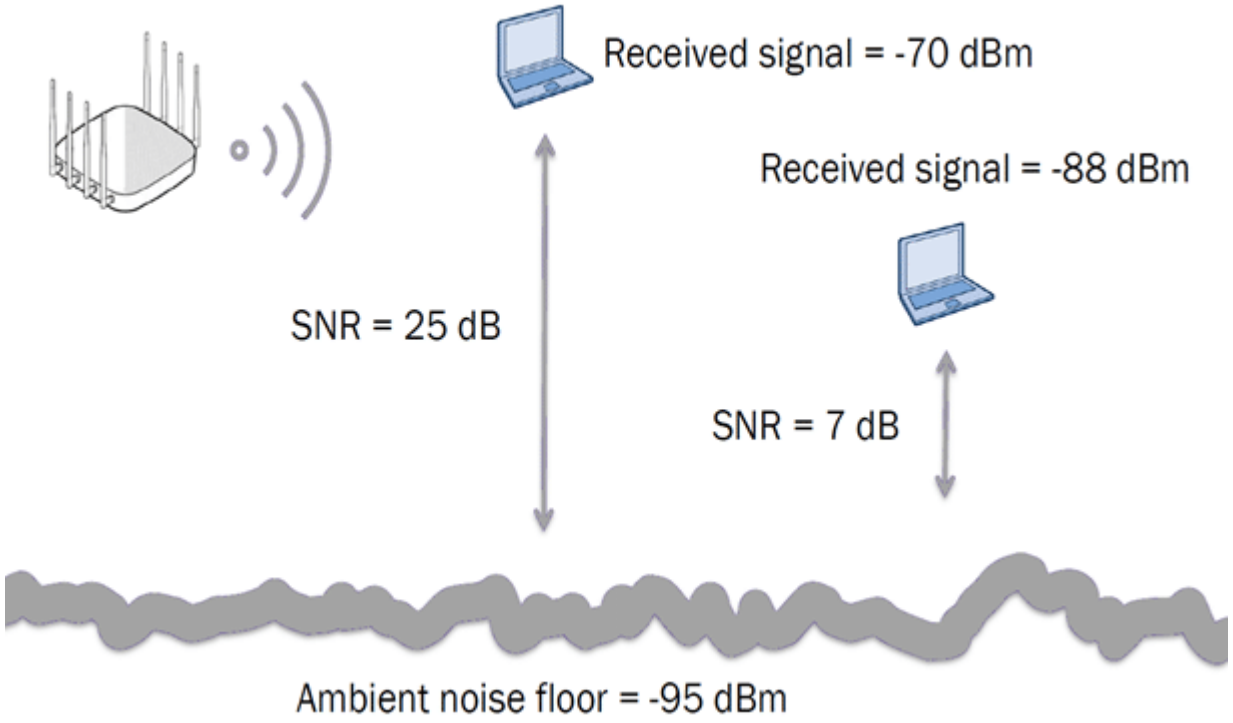
เทคนิคการเลือกใช้ช่อง Wi-Fi



Interference, RSSI, and SNR in Wi-Fi networks



Understanding RSSI and SNR in Wireless Networks for Effective Communication and Connectivity Improvement

Understanding RSSI and SNR

In a wireless LAN, devices transmit at power levels between 100 mW (20 dBm) and 1 mW (0 dBm). This transmission power, known as EIRP, is regulated by regional authorities. As the RF signal travels to the receiver, the received signal strength typically ranges from 0 dBm to about -100 dBm.

Understanding RSSI and SNR

This measurement is known as the **Received Signal Strength Indicator (RSSI)**. RSSI values indicate signal strength, with 0 dBm being the strongest and -100 dBm being the weakest. Each receiver has a sensitivity threshold. If the signal strength is above this threshold, the receiver can properly process the signal and extract useful information. However, if the signal strength falls below this level, the device cannot interpret the signal correctly.

Received Signal Strength

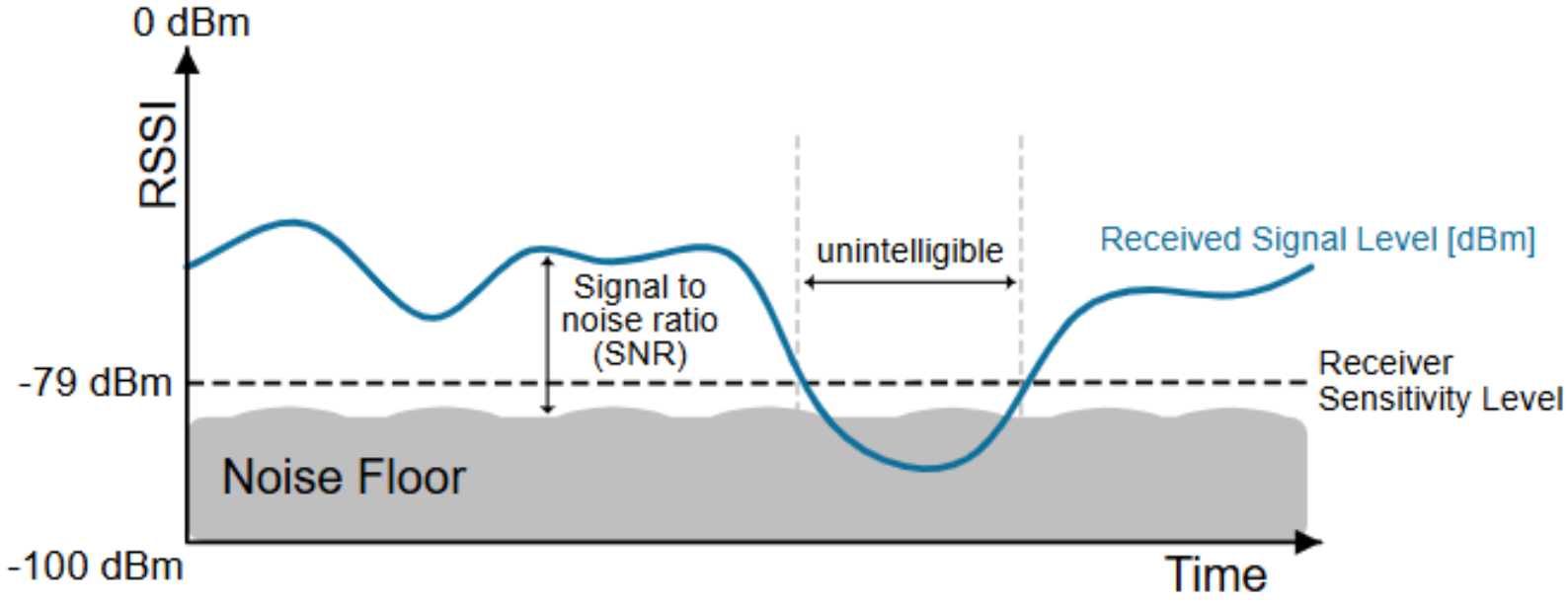


Figure 1. Received signal drops.

Received Signal Strength

Data Rate (Mbits/s)	Minimum Sensitivity (dBm)
6	-82
9	-81
12	-79
18	-77
24	-74
36	-70
48	-66
54	-65

Power supply	2*AAA battery , 3V
Wireless frequency	2.4GHz ~ 2.484GHz
Network protocol	IEEE802.11 b/g/n
Transmitting power	802.11b:14dBm±2dBm@1Mbps 802.11g:13.5dBm±2dBm@54Mbps 802.11n:13dBm±2dBm@MCS7_HT20
Receiving sensitivity	802.11b:-89dBm@11Mbps 8%PER 802.11g:-75dBm@54Mbps 10%PER 802.11n:-72dBm@MCS7_HT20 10%PER

Power (dBm)	Power (mW)
-50 dBm	.00001 mW
-60 dBm	.000001 mW
-70 dBm	.0000001 mW
-80 dBm	.00000001 mW
-90 dBm	.000000001 mW
-100 dBm	.0000000001 mW
-110 dBm	.00000000001 mW
-120 dBm	.000000000001 mW

For example:

A receiver may have a sensitivity threshold of -79 dBm.

If the received signal power level falls below this threshold, the signal becomes unintelligible, and the receiver cannot detect and interpret it, as shown in the diagram above.

Signal-to-Noise Ratio (SNR)

- A lower noise floor means that the receiver can detect weaker signals. We measure the strength of the received signal against the noise floor using the **Signal-to-Noise Ratio (SNR)**.
- It is expressed in decibels (dB) and calculated as:
- **SNR** = Received Signal Power (dBm) − Noise Power (dBm)

- **Good SNR**: An SNR above 20 dB is usually considered good. In this range, the signal is much clearer than the noise, leading to better communication quality.
- **Moderate SNR**: An SNR between 10 dB and 20 dB is considered moderate. The signal is still stronger than the noise, but there might be some noticeable interference.
- **Poor SNR**: An SNR below 10 dB is considered poor. In this range, the noise is almost as strong as the signal, making it difficult to understand the transmitted information clearly

- For example, the diagram below shows an RF signal with an average RSSI of -57 dBm. The noise floor averages at about -79 dBm.

  Therefore, we can calculate the SNR as:

  $$\text{SNR} = -57\ \text{dBm} - (-79\ \text{dBm}) = 22\ \text{dB}$$

- This is a good SNR value and the received signal. The receiver can detect the signal and demodulate the useful data.

Noise Floor Profile

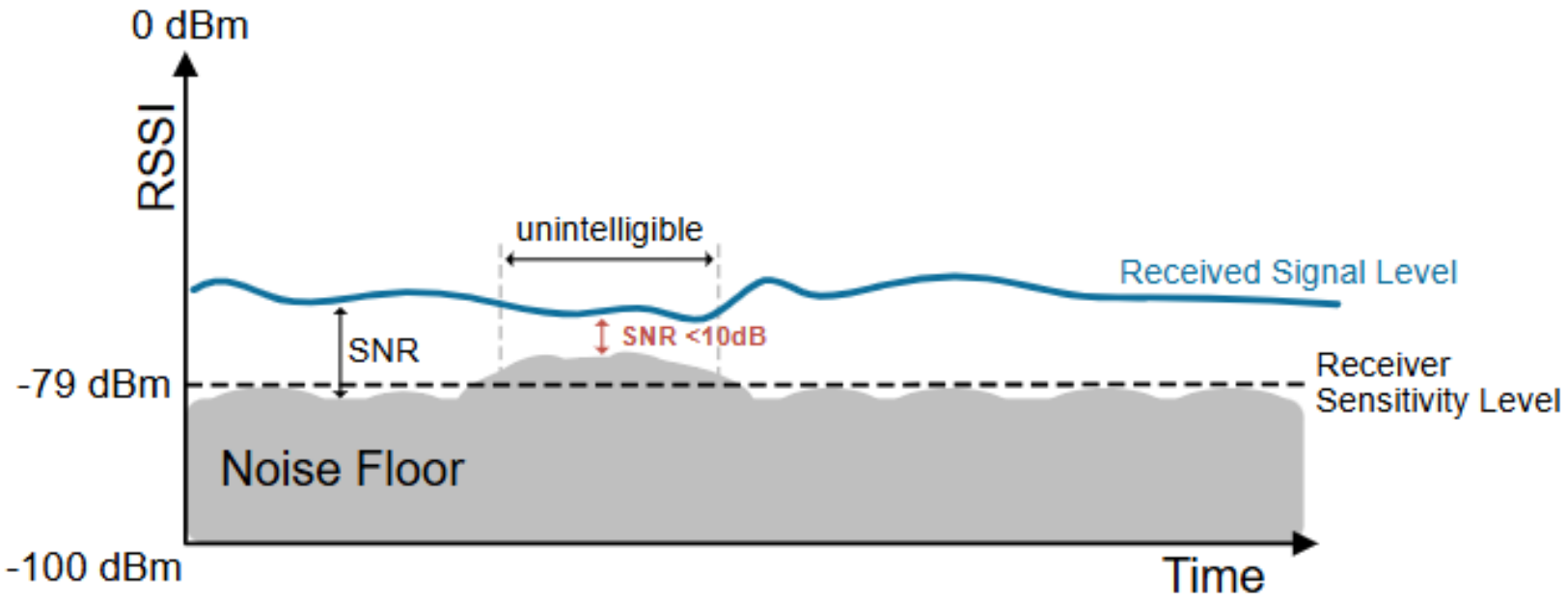


Figure 2. Noise Floor raises.

Understanding Wi-Fi Interference

- Interference in wireless communications occurs when unwanted signals disrupt the transmission and reception of data.

Co-channel interference (CCI)

- Co-channel interference (CCI) occurs **when multiple wireless devices use the same Wi-Fi channel** in the same place. For example, devices A and B transmit on the same channel 4 in the 2.4 GHz band. What happens is that their signals completely overlap throughout the entire 22-MHz width of the channel, as shown in the diagram below.

Channel Interference

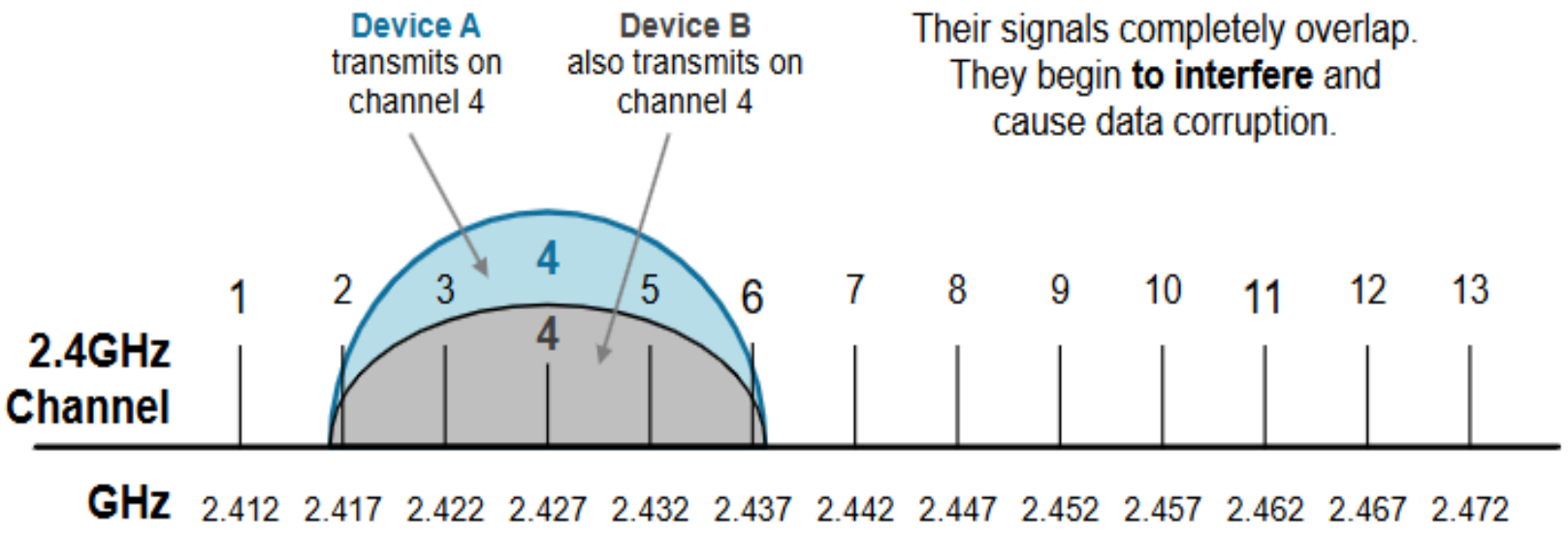
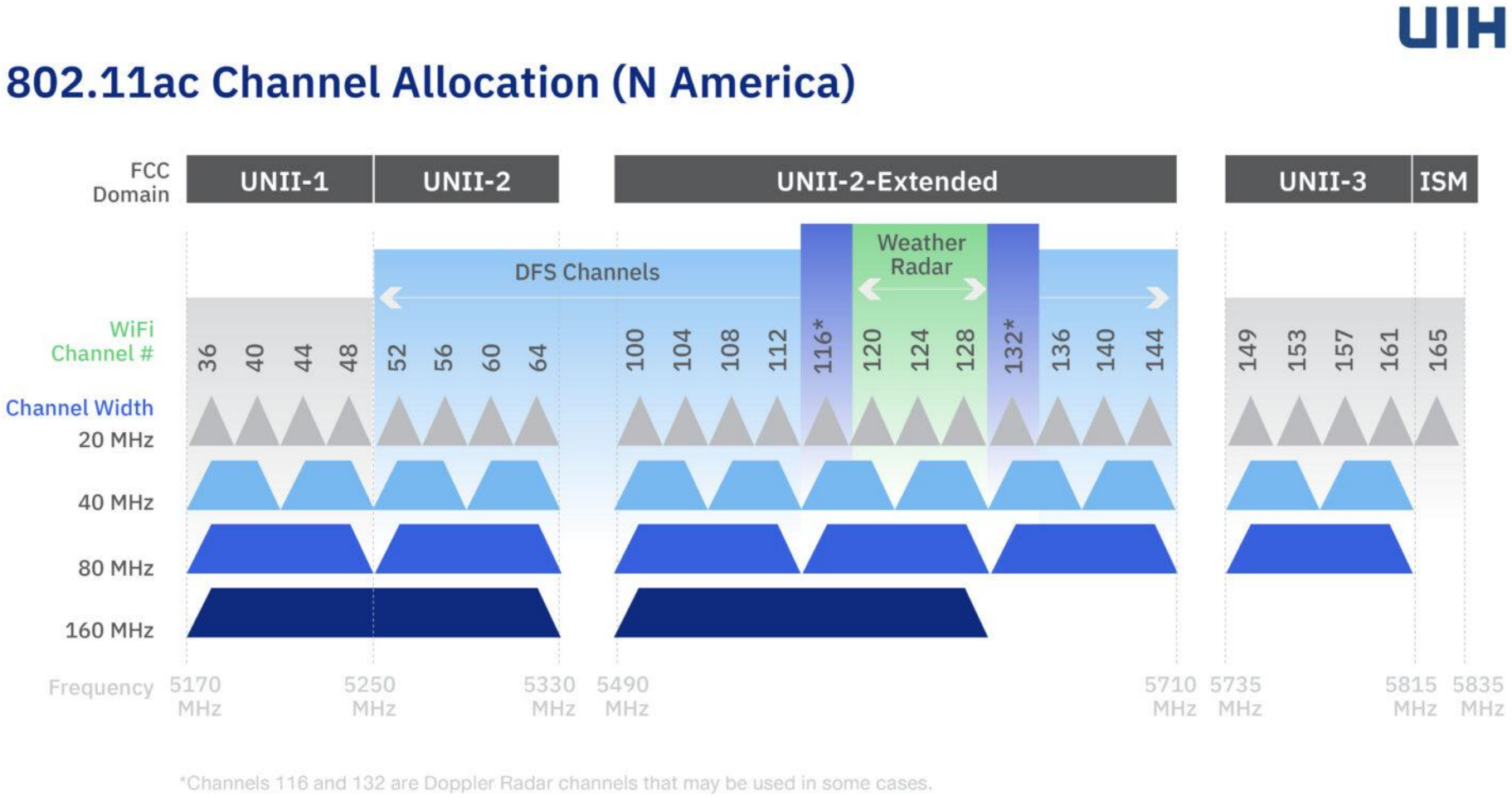


Figure 3. Same channel interference.

Channel Interference



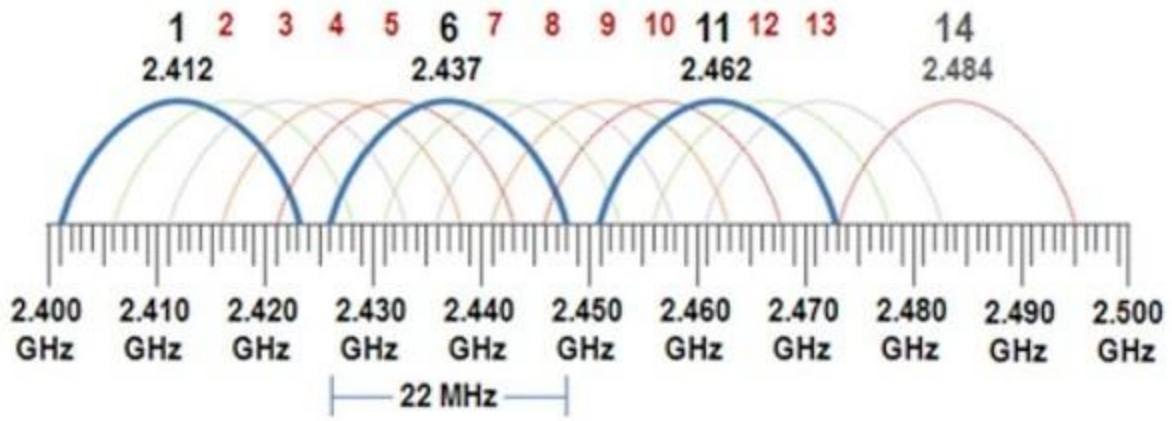
Channel Interference

2.4 GHz band lies between **2.400 GHz** and **2.4835 GHz**, but briefly called **2.4 GHz**.
5 GHz band lies between **5.150 GHz** and **5.825 GHz**, but briefly called **5 GHz**.

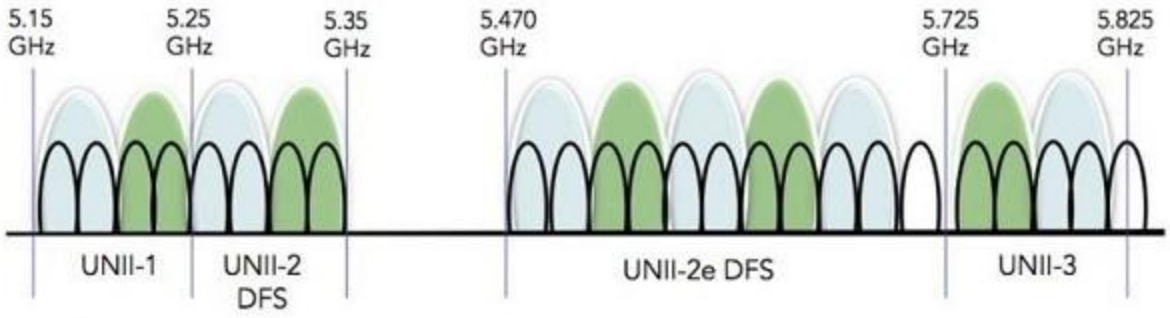
5 GHz band has **23 separate channels** that are not overlap. 5 GHz band has unoverlapping channels. So, using any channel with 5 GHz do not affect other Access Points.

2.4 GHz band has **14 channels** but each channels of 2.4 GHz is too wide, and each channel covers the frequency range of other four channels. So overlapping occurs. Using only **channel 1, 6** and **11** avoid overlapping.

2.4 GHz Bands



5 GHz Bands



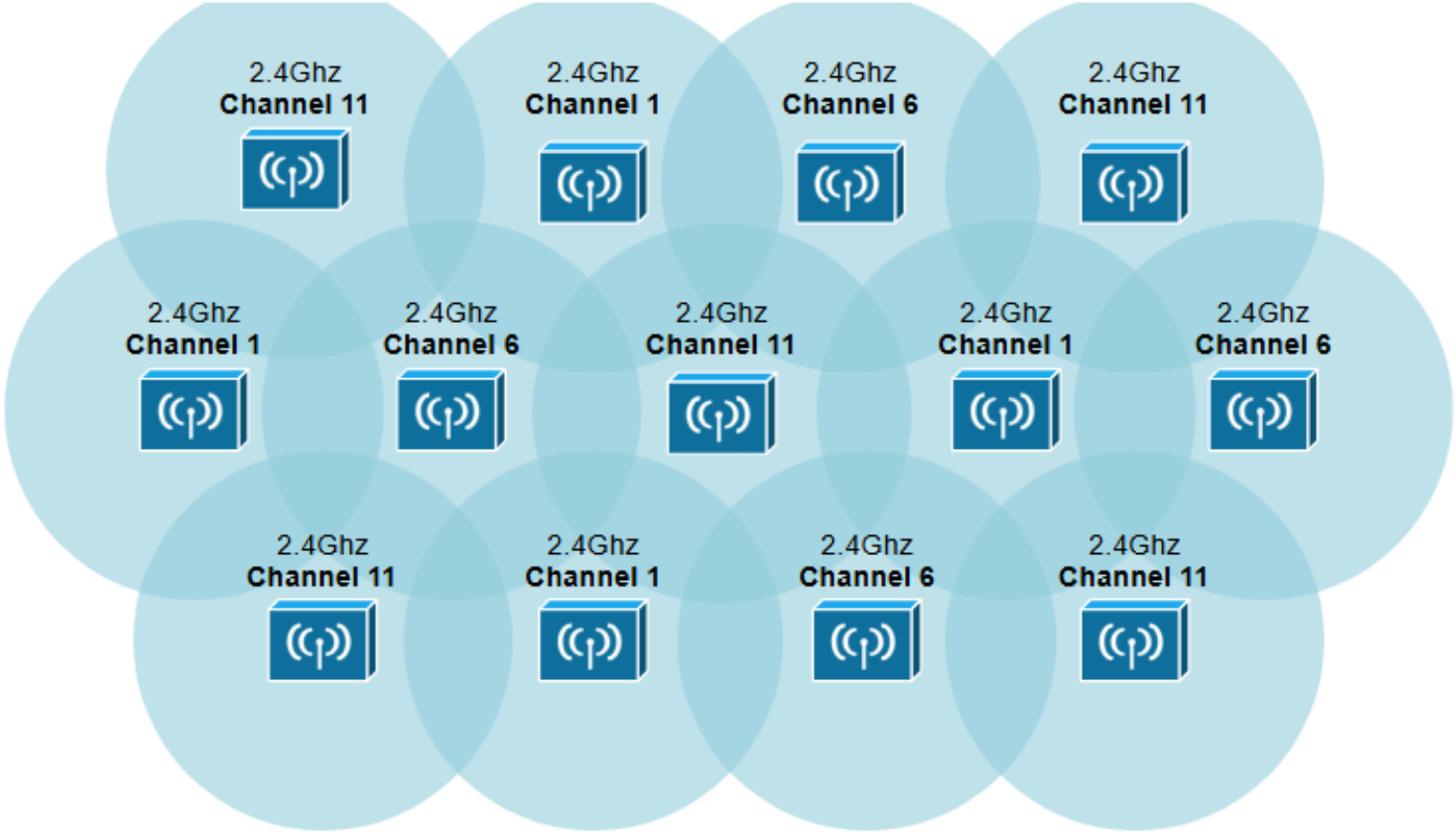


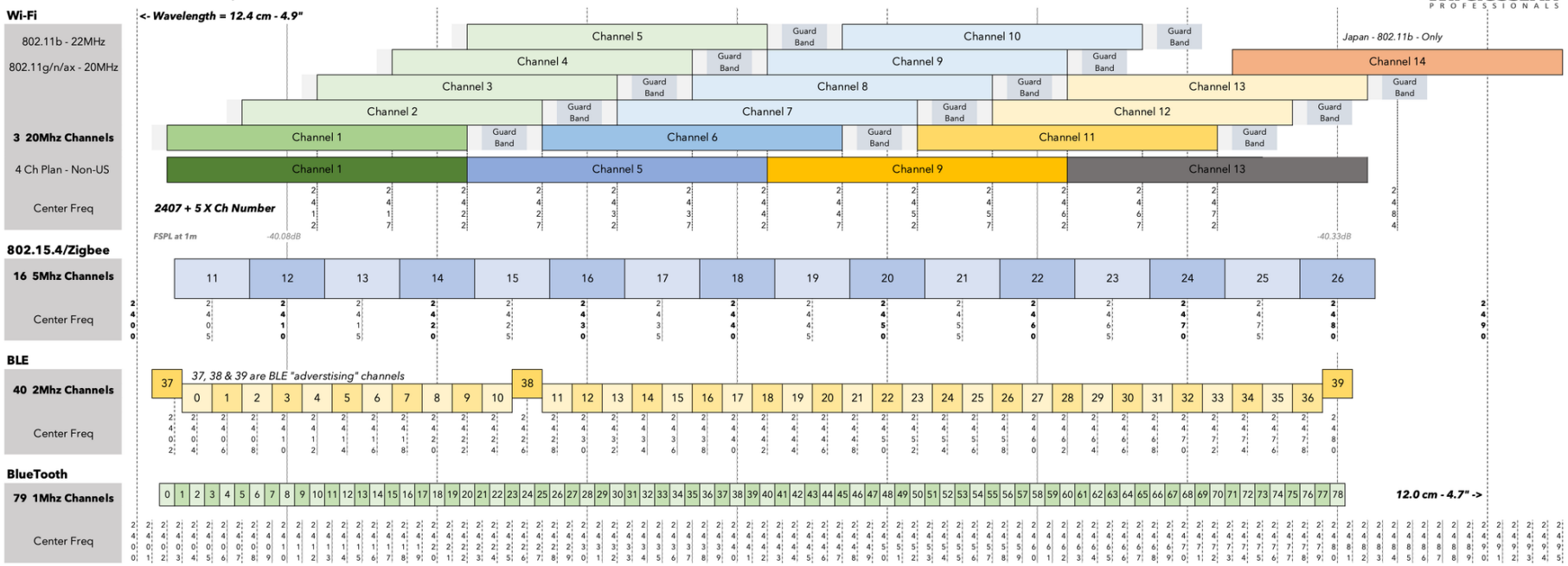
Figure 4. Channel Planning.

Wi-Fi คลื่นความถี่ 5 GHz มีแบนด์วิดท์ช่องสัญญาณ (Channels) 20 MHz

การแบ่งกลุ่มช่องสัญญาณ 5 GHz

- **UNII-1 (Low Band - ใช้งานในร่ม):** ช่อง 36, 40, 44, 48 (4 ช่อง)
- **UNII-2 (DFS - ใช้ร่วมกับเรดาร์ เช่น เรดาร์สภาพอากาศ):** ช่อง 52, 56, 60, 64 และช่อง 100 ถึง 140 (ประมาณ 12-13 ช่อง)
- **UNII-3 (High Band - ใช้งานทั่วไป/ภายนอก):** ช่อง 149, 153, 157, 161, 165 (5 ช่อง)

2.4GHz Unlicensed Spectrum



5 GHz Channel Allocations

500 MHz

| Qty | | |

- A good design takes advantage of wall attenuation. You configure an AP to use the same channel as another AP only if they are at least two walls away from each other. This provides enough separation so that the co-channel interference (CCI) is insignificant and both access points can maintain a healthy signal-to-noise ratio (SNR) in relation to each other, as shown in the diagram below.

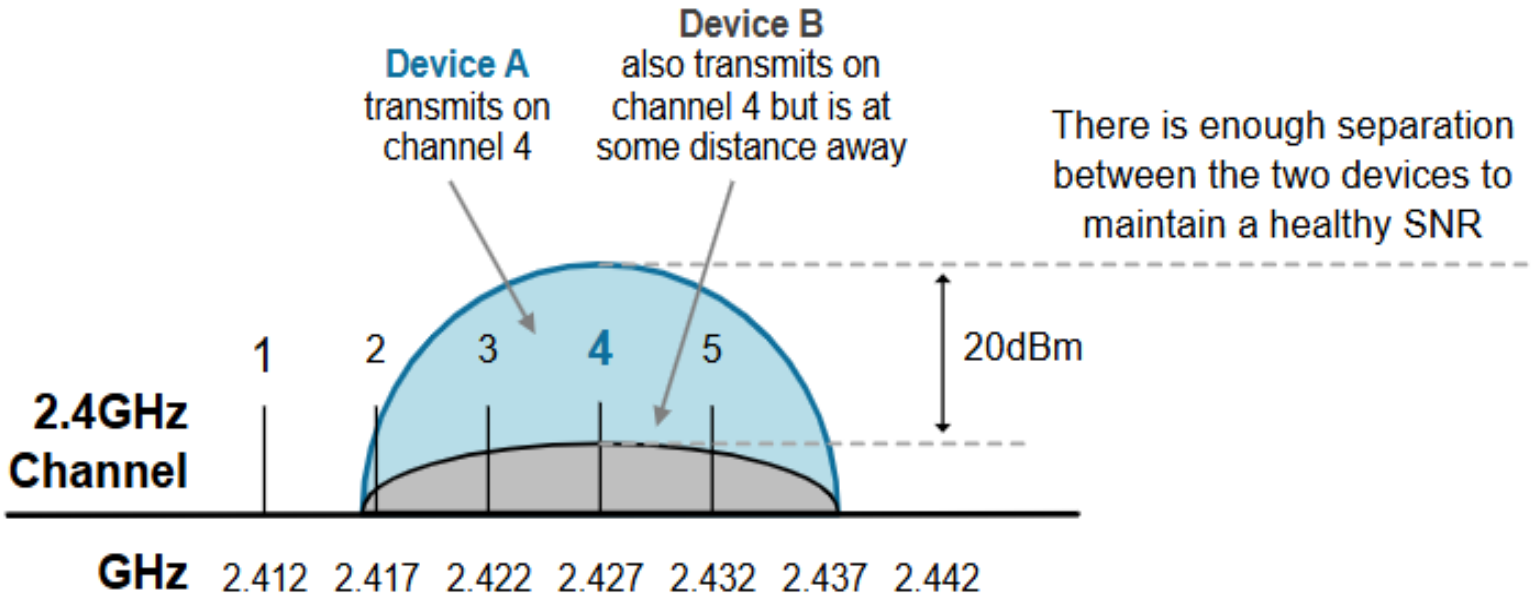


Figure 5. Minimizing CCI interference.

- However, wireless devices can interfere even if they are not transmitting on the same WiFi channel. This is called neighboring channel interference.

Neighboring channel interference (NCI)

- Neighboring Channel Interference (NCI) happens when two wireless signals **on nearby frequencies** interfere with each other. This usually occurs when Wi-Fi devices operate on channels that are too close together.

Recall that a WiFi channel has a bandwidth of 22 MHz

This means that if a device transmits on channel 4, for example, it uses frequencies between 2.416 and 2.438 GHz and not only the central channel frequency of 2.427 GHz. Therefore, channels can interfere with each other, as shown in the diagram below.

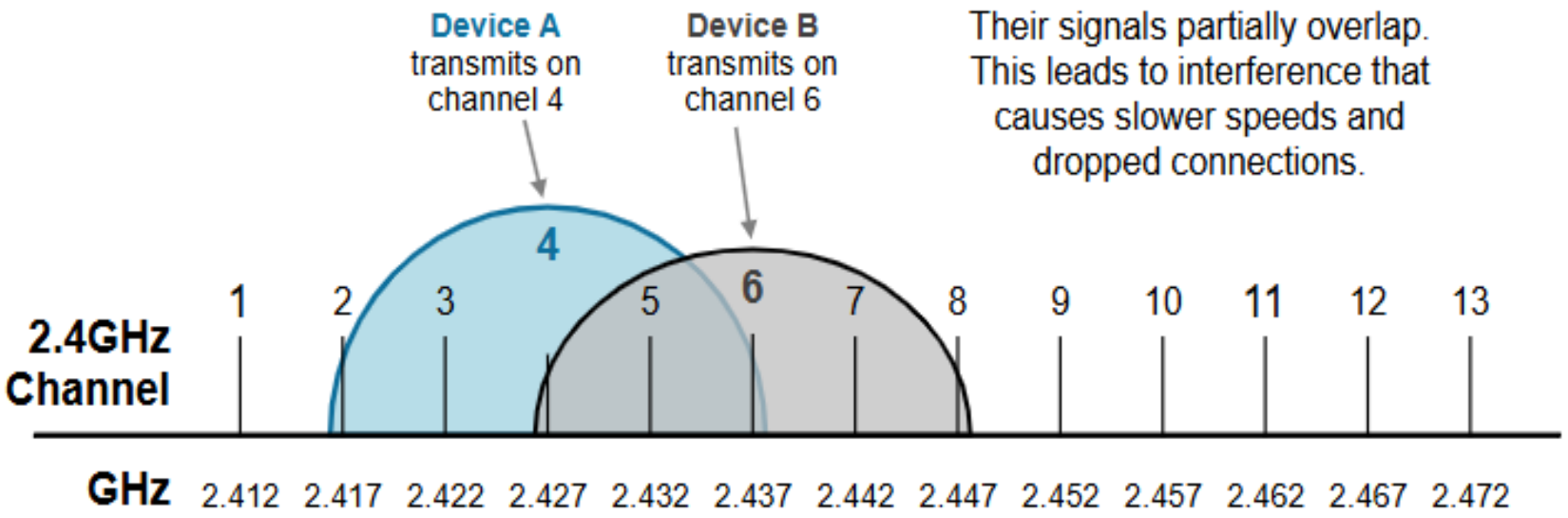


Figure 6. Neighboring channel interference.

- To avoid NCI, it's best to use non-overlapping channels. The best channels to use in the 2.4 GHz Wi-Fi band are **1, 6, and 11 because they do not overlap**. More channels are available in the 5 GHz band, providing more flexibility in designing the wireless LAN so that NCI does not occur. However, a good design may require more access points (AP) configured to transmit at lower power levels.

- There could be many sources of non-Wi-Fi interference in every frequency band (2.4 GHz, 5 GHz, and 6 GHz). The most common ones include:
  - Microwaves
  - Bluetooth devices
  - Cordless phones
  - Baby monitors
  - Wireless cameras
  - Radar systems (especially in 5 GHz) etc.

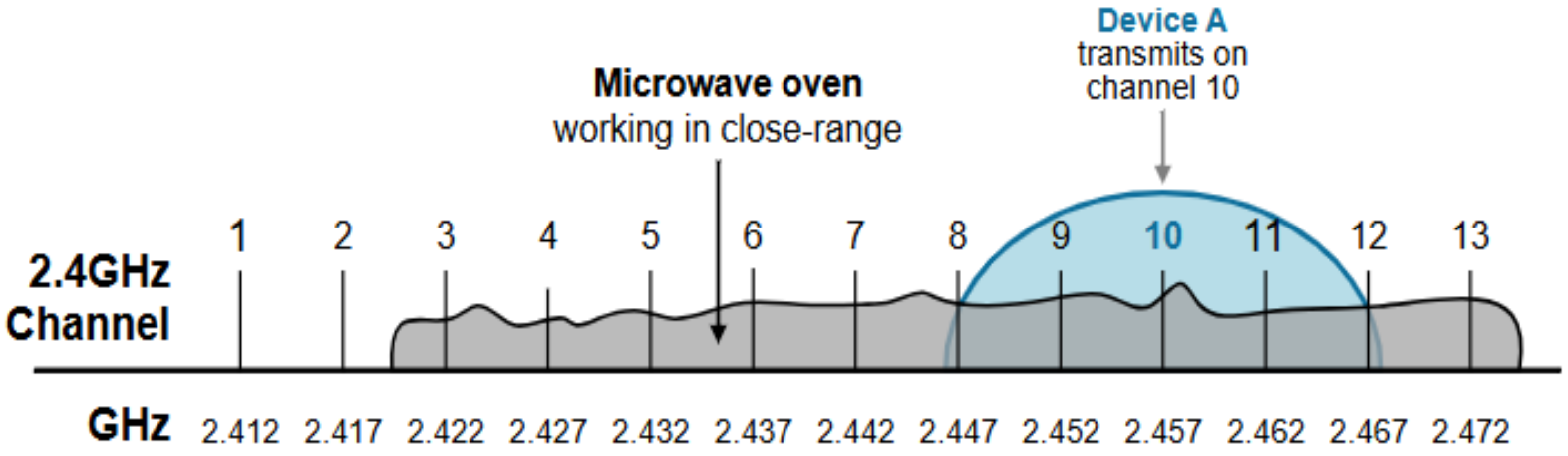


Figure 7. Non Wi-Fi interference.

Free Space Path loss (FSPL)

When a radio frequency (RF) signal is transmitted from an antenna, its strength decreases as it moves through open space. Even with no obstacles in the way, the signal still weakens.

This effect is called *free space path loss (FSPL)*.

The loss is relative to two RF signal components: Frequency and Distance.

The free space path loss (in decibels) is calculated using the formula:

$$\mathbf{FSPL(dB) = 20log10(d) + 20log10(f) + 92.45}$$

where:

d = distance in kilometers (km)

f = frequency in gigahertz (GHz) and

92.45 is a constant adjusted for GHz and km units.

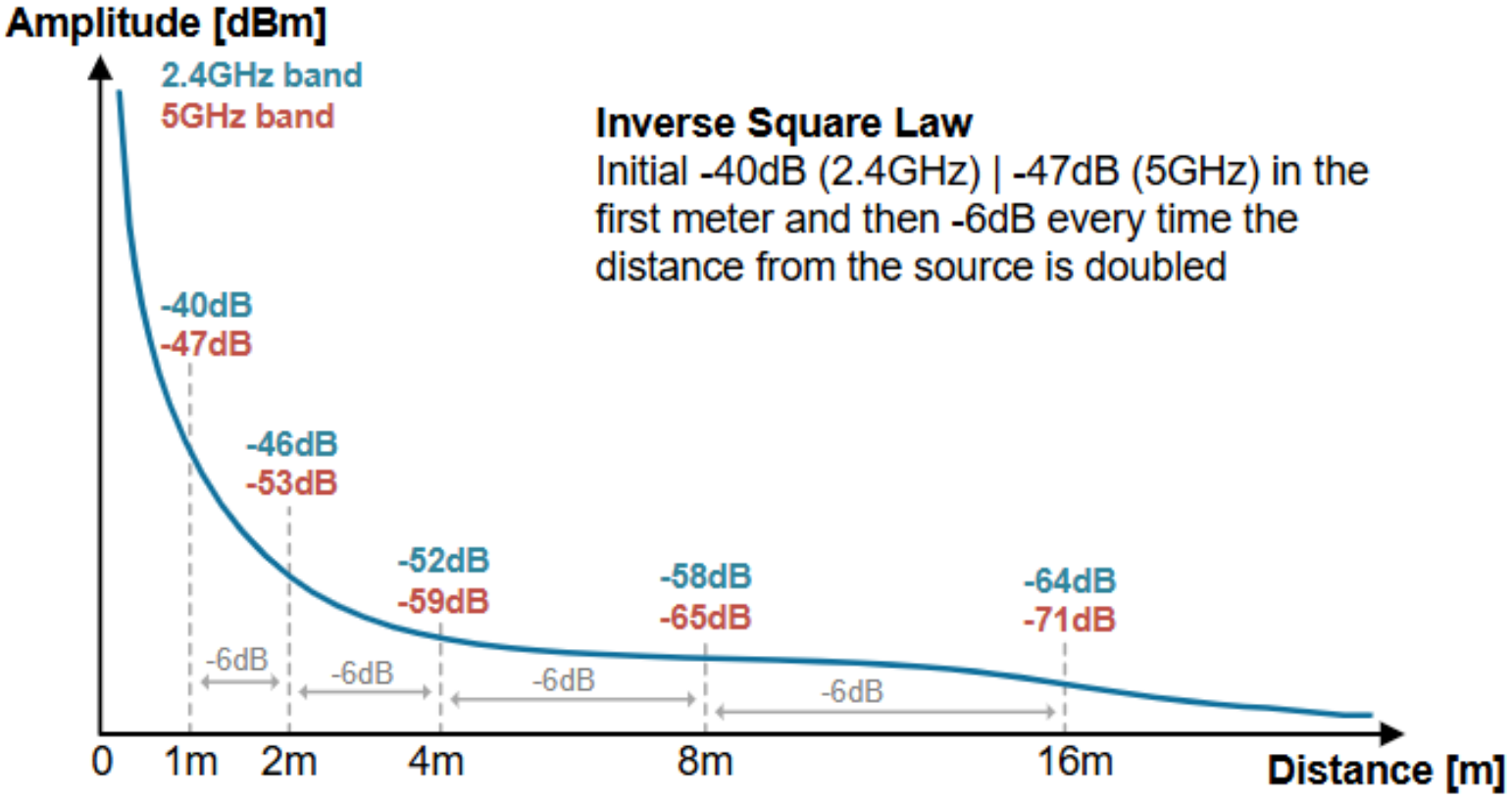


Figure 8. Free Space Path loss.

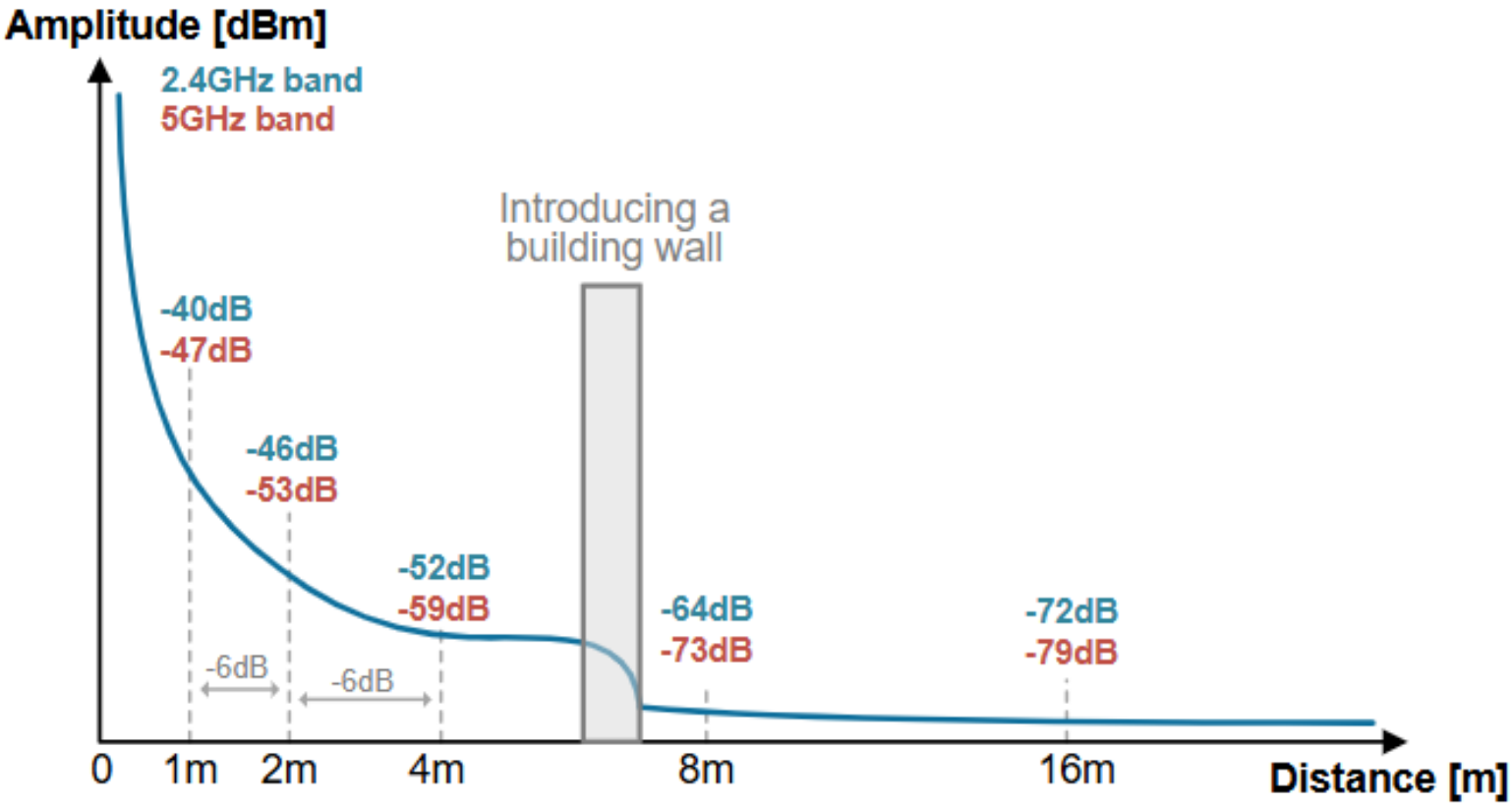


Figure 9 FSPL plus a wall.

Material	Attenuation @ 900 MHz
Glass 0.25" (6 mm)	0.8 dB
Glass 0.5" (13 mm)	2 dB
Lumber 3" (76 mm)	2.8 dB
Brick 3.5" (89 mm)	3.5 dB
Brick 7" (178 mm)	5 dB
Brick 10.5" (267 mm)	7 dB
Concrete 4" (102 mm)	12 dB
Masonry Block 8" (203 mm)	12 dB
Brick faced concrete 7.5 " (192 mm)	14 dB
Masonry Block 16" (406 mm)	17 dB
Concrete 8" (203 mm)	23 dB
Reinforced Concrete 3.5" (203 mm)	27 dB
Masonry Block 24" (610 mm)	28 dB
Concrete 12" (305 mm)	35 dB

Table 2.

- Wi-Fi devices **automatically adjust their modulation and coding based on signal strength**. If the signal is strong, they use complex methods to send more data at higher speeds. If the signal weakens, they switch to simpler methods to maintain a stable connection, even if it means slower speeds. This process is called *dynamic rate shifting (DRS)* and happens automatically without user intervention.

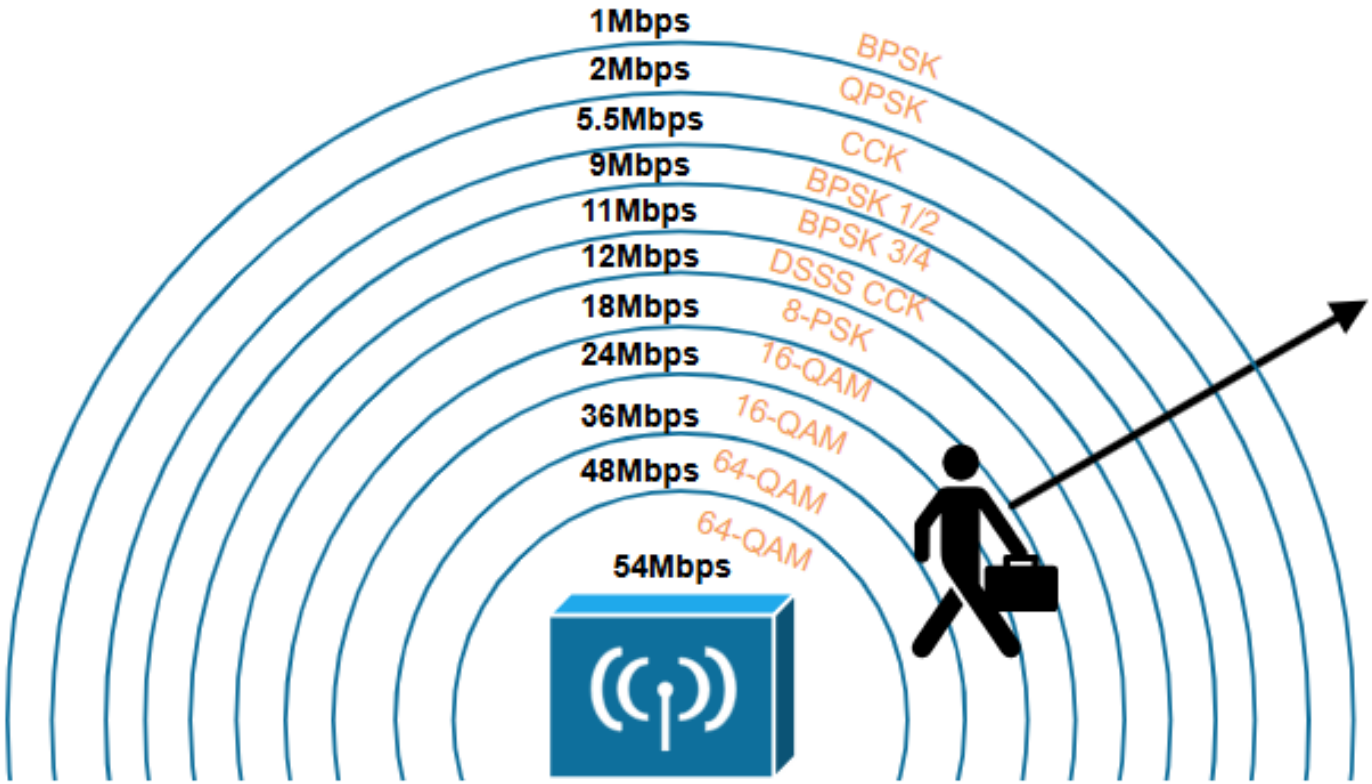
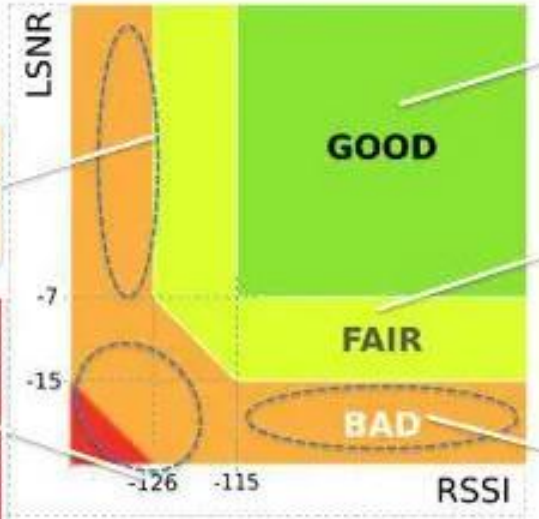


Figure 10. Dynamic rate shifting.

Radio level analysis

Reference test (line of sight) :

- dist. 1m → RSSI – 30 dB
- dist. 2m → RSSI – 40 dB
- ✓ Dist. x2 → RSSI – 6dB



FAR AWAY environment
→Try to put device higher

NO or INSUFFICIENT reception
→ Try to put the gateway higher or maybe you will have to install a additional gateway
→ In all cases, a new Tests session must be done

RF level is optimal to get a good reception reliability

RF level is not optimal but must be sufficient
→ Try to improve your device position if possible
→ You will have to monitor the stability of the RF level

NOISY environment
→Try to put device out of electromagnetic sources

3

RSSI: Received Signal Strength Indicator
LSNR: LoRa Signal Noise Rate

RULES:

- ❑ Sensor at least at 1 meters of the gateway antenna
- ❑ It always depends of the global environment

References

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https://ipcisco.com/lesson/wlan-frequency-bands/
https://community.ui.com/questions/Understanding-Channels-and-Banding/b89cde23-a018-4b67-8aff-65f150bd7d29